



Faculty of Information Technology

Semester 2, 2026

FIT5145: Foundations of Data Science

Assignment 4: Description

Friday, October 30, 2026, at 11:55 PM

Assessment Details:

- Assessment Type: Individual Assignment
- Total marks (40%): Written coding assignment (25%) + Online interview (15%)
- Due Date: Friday, October 30, 2026, at 11:55 PM. Please note that we do not accept submissions after November 6, 2026 (i.e., 7 days after the due date).
- **Online interview:** The online interview will be approximately 10 minutes long and will be held in Week 14. The exact interview time for each student will be announced on the Ed Forum prior to the interview.

Submission Details:

For this assignment, 4 files should be submitted: **two PDF reports, one RMD file, and one CSV file.**

1. Task A PDF report:

- For Task A, make sure to include all *(a) shell code, (b) code output and answer, and (c) explanation* for each question of Task A.

(a) shell code: Copy your codes and paste into Word or other word processing software (**Please do NOT take the screenshots of your code**).

(b) code output: Include screenshots of the code outputs.

(c) explanation: Explain your codes or summarise your work. Include justification for your approach where appropriate.

2. Task B-C PDF report: Make sure to include all *(a) R code, (b) code output and answer, and (c) explanation* for each question of Tasks B and C.

Please convert (knit) the RMD file to HTML, print the HTML (Select 'Save as PDF'). Please ensure that your PDF displays all code, outputs, and explanations.

(a) R code: Use a separate code chunk for each question or task.

(b) code output: Include the code outputs.

(c) explanation: Explain your codes or summarise your work. Include justification for your approach where appropriate.

3. **Task B-C RMD file:** This is the RMD file used to produce the Task B-C PDF report, which means the Task B-C RMD file is identical to the Task B-C PDF report.
4. **CSV file** containing the **predicted** labels required for Task C

Notes:

1. Whenever you identify anomalies, errors, or any issues in the dataset that may negatively affect the validity of your analysis, **you are expected to determine and justify an appropriate approach and perform the necessary data wrangling.**

This assignment also assesses your ability to critically evaluate data quality and make informed preprocessing decisions. Different approaches may be acceptable, provided they are clearly justified and appropriate for the analysis.

2. Whenever a question asks for a certain value, your code should produce the value. For example, when a question asks for the number of rows contained in a table, your code should print out the number of rows. **Extraction of the answer manually by eye-examination will not earn any marks.**
3. Please note that you are only allowed to use shell commands for Task A and R for Tasks B and C.
4. **Please make sure that you can select and highlight texts in your PDF**, as shown below then the turnitin score can be generated properly for your PDF file (we just need the Turnitin score for the PDF file, not the RMD file).



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1. Task A PDF report:

- For Task A, make sure to include all (a) shell code, (b) code output and answer,

Task A: Shell commands

You work in the data team at a consumer lender. Your manager sends you the data file “*consumer_complaints.csv*” provided by [Consumer Financial Protection Bureau](#) and wants you to analyse it and extract insights to support management's decisions and policy. The file contains consumer complaints about financial products and services to companies, and companies’ responses to the complaints. In the file, there are different variables to describe each complaint about the products and services, as described below. Please refer to this [link](#) to get more information about the data. Please download “*consumer_complaints.csv*” from Moodle.

Column Name	Description
<i>Complaint ID</i>	The unique identification number for a complaint
<i>Date_received</i>	The date of the complaint received

<i>Product</i>	The type of product the consumer identified in the complaint
<i>Sub-product</i>	The type of sub-product the consumer identified in the complaint
<i>Issue</i>	The issue the consumer identified in the complaint
<i>Sub-issue</i>	The sub-issue the consumer identified in the complaint
<i>Consumer complaint narrative</i>	Consumer complaint narrative is the consumer-submitted description of "what happened" from the complaint
<i>Company public response</i>	Companies can choose to select a response from a pre-set list of options that will be posted on the public database
<i>Company</i>	The complaint is about this company
<i>State</i>	The state of the mailing address provided by the consumer
<i>ZIP code</i>	The mailing ZIP code provided by the consumer
<i>Tags</i>	Data that supports easier searching and sorting of complaints submitted by or on behalf of consumers.
<i>Consumer consent provided?</i>	Identifies whether the consumer opted in to publish their complaint narrative.
<i>Submitted via</i>	How the complaint was submitted

Please note that **you are only allowed to use shell commands** for Task A, as you would run in Linux shell, Mac terminal, or Cygwin, to tackle this task. Using other utilities or tools such as PowerShell is NOT allowed.

Your manager has asked you to answer the following questions:

1. What is the *Date_received* range of the collected complaints?
2. We want to preprocess the *Complaint_ID* and *Date_received* columns.
 - a. **Count** lines with a *complaint id* that is not a number of 7 digits long, i.e., *id* values that contain anything other than numbers OR are of a length more/less than 7.
 - b. **Remove** the lines mentioned in Q2-a and **remove** time values in the *Date_received* column. For example, the *Date_received* column will contain "29/04/2020", instead of having "29/04/2020 23:13".
 - c. **Display** the first 3 lines (including a header) of the dataset that was filtered in Question 2-b. Store the filtered dataset in a file named "***filtered_complaints.csv***" and **use this file for the remaining questions in Task A.**
3. A manager wants to investigate the top five products with the most complaints.
 - a. How many product types are there in the dataset?

- b. What are the top 5 most frequent products in the dataset (i.e., the top 5 products with the largest number of complaints)?
4. A manager would like to better understand common issues reported by customers. Let's investigate the *Consumer complaint narrative* column. The manager wants to know the following:
 - a. How many complaints are related to credit card fraud? (Note: Please ignore cases)
 - b. How many complaints mention long wait times? (Note: Please ignore cases and include the length of time waited.)

Task B: Exploratory Data Analysis Using R

You work as a data scientist at a real estate company in Melbourne. Rents and home prices have risen sharply over the past year, and your manager wants to understand what the transaction data actually shows before the company advises its clients. She asks you to perform exploratory data analysis on “*property_transaction_victoria.csv*”, which contains most of the property transactions that took place in Greater Melbourne between 2010 and 2023. The data was collected from one of the top real estate websites. The file contains different variables describing each collected transaction record, as described below. Please download “*property_transaction_victoria.csv*” from Moodle.

Column Name	Description
<i>id</i>	The unique ID of a transaction record, which usually consists of 9 digits
<i>badge</i>	Whether a property is for rent or buy or it's already sold
<i>url</i>	URL of a property
<i>suburb</i>	The suburb where a property locates in
<i>state</i>	The state where a property locates in
<i>postcode</i>	The postcode of a property
<i>short_address</i>	Short address of the property
<i>full_address</i>	Full address of the property
<i>property_type</i>	Whether a property is a <i>House</i> , <i>Townhouse</i> , etc.
<i>price</i>	The price for which the property was sold
<i>bedrooms</i>	The number of bedrooms that a property has

<i>bathrooms</i>	The number of bathrooms that a property has
<i>parking_spaces</i>	The number of parking spaces that a property has
<i>building_size</i>	The building size of a property
<i>building_size_unit</i>	The unit of building size of a property (measured in square metres)
<i>land_size</i>	The area size of a property
<i>land_size_unit</i>	The unit of the area size of a property (measured in square metres)
<i>listing_company_id</i>	Real estate agent who managed the transaction
<i>listing_company_name</i>	Name of a real estate agent
<i>listing_company_phone</i>	Phone number of a real estate agent
<i>auction_date</i>	Date of a property auction
<i>available_date</i>	Available date that a buyer can move into a property
<i>sold_date</i>	The date on which the transaction was made
<i>description</i>	A textual description that real estate agents used to describe the property and attract potential buyers before the transaction was made.
<i>images</i>	Url of a property images
<i>images_floorplans</i>	Url of a property floor plan images
<i>listers</i>	List of the real estate agent information
<i>inspections</i>	Inspection date on a property

1. Identify the top 3 suburbs with the highest number of property transactions over the years, and plot their monthly transaction counts for the year 2022. Include Toorak in the plot as well, if it is not among the top 3 suburbs.
2. What are the 3 most important keywords in the description column that impact property prices? (Note: Since the description column contains a large volume of text, please extract a 10% sample from the original dataset to answer this question.)
3. Owning property has long been considered a reliable way to build personal wealth. Which properties have experienced the highest price increases since their first sale? Please exclude properties where the time between the first and last sale exceeds five years. List the top five properties along with their address, capital gain, and the

duration between the first and last sale.

4. Chris is looking for a renovated house to purchase. He wants the property to be close to a shopping centre, and since he has an 8-year-old son, proximity to a primary school is also important. He is looking for a home with 4 bedrooms and 2 bathrooms. Currently, he is considering six suburbs - Mulgrave, Vermont South, Doncaster East, Rowville, Glen Waverley, and Wheelers Hill and plans to choose one from this list. Please provide the predicted price for November 2026 of a house that meets the above criteria in each of the six suburbs, and display the predicted price alongside the corresponding suburb name. (Note: When you build the prediction model, please use only the provided dataset and the period it covers.)

Task C: Predictive Data Analysis Using R

People's verbal and written language (e.g.: blog, Ed forum on Moodle) can provide rich information about their beliefs, fears, thinking patterns, social relationships, and personalities. Text analysis has been widely used to analyse natural language (text) and gain insights, and there have been many studies in the field of text analysis. We introduce one of approaches, LIWC (Linguistic Inquiry and Word Count), which extracts the various emotional, cognitive, and structural components from language. You can also find more detailed information about LIWC in the 'Manual_LIWC.pdf' attached on Moodle and on the [LIWC website](#).

In this task, you are required to analyse student posts from Moodle discussion forums in Monash University units. The objective is to apply machine learning models to classify posts into two categories:

- Content-relevant: posts related to the knowledge taught in the course
- Content-irrelevant: posts not directly related to course content

A set of features (primarily engineered using LIWC) has been provided for each forum post to support model training. Your goal is to build machine learning models that predict whether a post is related to unit content (i.e., the values in the *label* column).

- Independent variables: Columns B–CR in the datasets (from *WC* to *demographic_sex*)
- Dependent variable: The *label* column

The following datasets are available:

- Training: forum_ind_train.csv, forum_dep_train.csv
- Validation: forum_ind_validation.csv, forum_dep_validation.csv

- Testing: forum_ind_test.csv, forum_dep_test.csv

All datasets can be downloaded from Moodle. A detailed feature description table is provided below, outlining the meaning of each column in the files.

Column ID	Column name (Column index)	Description
1	Unique_ID (Column A)	Unique ID of a forum post selected from the discussion forum of a unit at Monash University
2	Variables (Columns B-CP)	Variables generated by applying LIWC, which is a transparent text analysis program that counts words in psychologically meaningful categories. The detailed explanations of the variables can be found in Table 4 of the PDF document “ <i>Manual_LIWC.pdf</i> ” on Moodle.
3	unit_faculty (Column CQ)	Faculty from which the unit belongs to
4	demographic_sex (Column CR)	Sex information of the student who made the forum post
5	label	Label to be predicted for the post. There are two classes in the label, i.e., content-relevant (being related to the unit content, e.g., "what is data management") and content-irrelevant (not being related to the unit content).

Note: Values in the features (Columns B-CP) have been generated by the LIWC program and you can interpret them as for larger feature values, they appear more frequently in the text.

1. Please build multiple predictive models using methods you consider appropriate. Train each model on the training set and evaluate its performance on the validation set using the [F1 score](#). Compare the models and identify which one performs best.

Your goal is not simply to fit models, but to improve predictive performance through a well-reasoned modelling process. It is up to you to decide what steps are necessary to get the most out of the data and the algorithms you choose.

For every decision you make, explain what you did, why you did it, and whether it improved performance (with evidence). Your submission should include relevant R code, outputs, and model performance results. Marks will be awarded based on the depth of your investigation, the quality and appropriateness of your modelling process, and the strength of the justification for the choices you made.

2. Your model's performance depends on how complex you allow it to be. Investigate this relationship for your chosen model and present a visualisation as evidence that lets you reason about underfitting, overfitting, and the bias-variance trade-off. Use what you find to justify the complexity of your final model.

3. Please notice that the groundtruth *label* values in the file “*forum_dep_test.csv*” are withheld for now, but they will be shared after the due date of this assignment. Here, your task is to use the best-performing model constructed from Question 1 to predict the label of the posts contained in the file “*forum_dep_test.csv*”. You need to populate your prediction results (i.e., the predicted *label* values) into the file “*forum_dep_test.csv*” and upload it to Moodle to measure the overall performance of your model. Please ensure the number of columns and rows and column names remains the same as in the original file (in the file “*forum_dep_test.csv*”), and only fill in the prediction results in the '*label*' column. Please name the submission file using the following format:

LastName_StudentNumber_forum_label_test.csv.

The mark you receive for this question will be dependent on the performance level of your model (measured by F1 score).